

Optimizing MeraDoc's Kubernetes Infrastructure on AWS EKS with Spot Instances & KubeCost

Client Overview by MeraDoc

MeraDoc is a tech-enabled healthcare delivery platform offering appointment booking, elderly care, and holistic health management. As the platform scaled, optimizing cloud infrastructure while ensuring **zero downtime** became critical.

Challenges

High AWS cloud costs due to **on-demand EC2 instances**.
Need for a **scalable, resilient** Kubernetes architecture.
Ensuring **100% uptime** while reducing cloud expenses.
Lack of visibility into **resource utilization and spending**.

Our Solution

OnEggy Technologies re-architected MeraDoc's **AWS EKS (Elastic Kubernetes Service) setup** with a cost-effective and performance-driven approach:

Migrated compute workloads to AWS Spot Instances, reducing costs by up to **40%**.

- ✓ Implemented **Auto-Scaling with Node Groups**, ensuring uninterrupted service even if spot instances were reclaimed.
- ✓ Integrated **KubeCost** to monitor and optimize resource allocation, identifying over-provisioned workloads.
- ✓ Used **Cluster Autoscaler** to dynamically adjust node capacity based on real-time demand.
- ✓ Enhanced **observability with Prometheus & Grafana**, tracking performance and cost trends.

Results & Impact

40% reduction in AWS costs while maintaining **99.99% uptime**.

Zero downtime migration to spot-based workloads using preemptive instance handling.

Real-time cost visibility with KubeCost, leading to optimized resource allocation.

Automated scaling for high-traffic scenarios, ensuring uninterrupted healthcare service delivery.

Tech Stack Used

AWS EKS | Terraform | Karpenter | KubeCost | Prometheus | Grafana | Loki | Jenkins | Nginx Ingress

Final Thoughts

By leveraging **spot instances, auto-scaling, and cost monitoring**, MeraDoc now enjoys a **cost-efficient, scalable, and high-availability Kubernetes infrastructure**—enabling seamless healthcare services.

Looking to optimize your Kubernetes infrastructure without compromising uptime?

Let's talk!